

Customer Shopping Behavior Analysis

Analyzing **3,900 transactions** across product categories to uncover spending patterns, customer segments, and strategic insights using Python, SQL, and Power BI.



Dataset at a Glance

3,900

Total Purchases

Rows in dataset

18

Features

Columns tracked

37

Missing Values

Review Rating column only

\$59.76

Avg. Purchase

Mean transaction value

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount Applied, Previous Purchases, Review Rating, Shipping Type).

Data Preparation in Python

Cleaning & Standardization

Loaded data via pandas, checked structure with `df.info()` and `.describe()`, imputed missing Review Ratings using median per category, and renamed columns to snake_case.

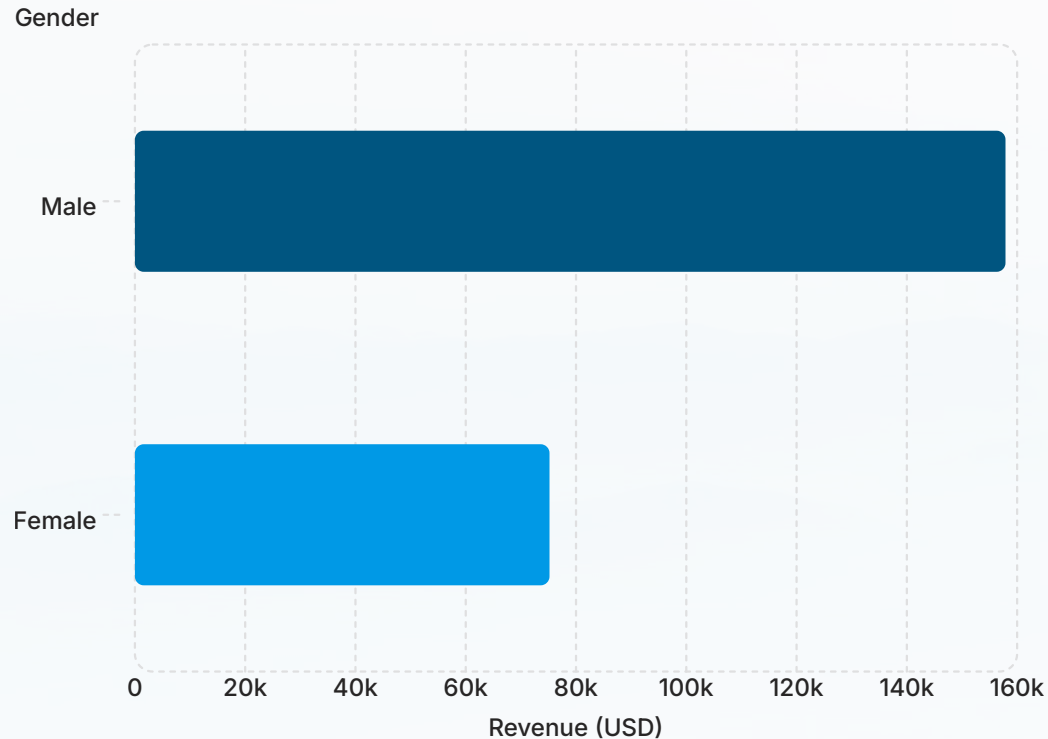
Feature Engineering

Created `age_group` by binning customer ages and `purchase_frequency_days` from purchase data. Dropped redundant `promo_code_used` (duplicate of `discount_applied`).

Database Integration

Connected Python to PostgreSQL and loaded the cleaned DataFrame for SQL analysis.

Revenue & Spending Insights (SQL)

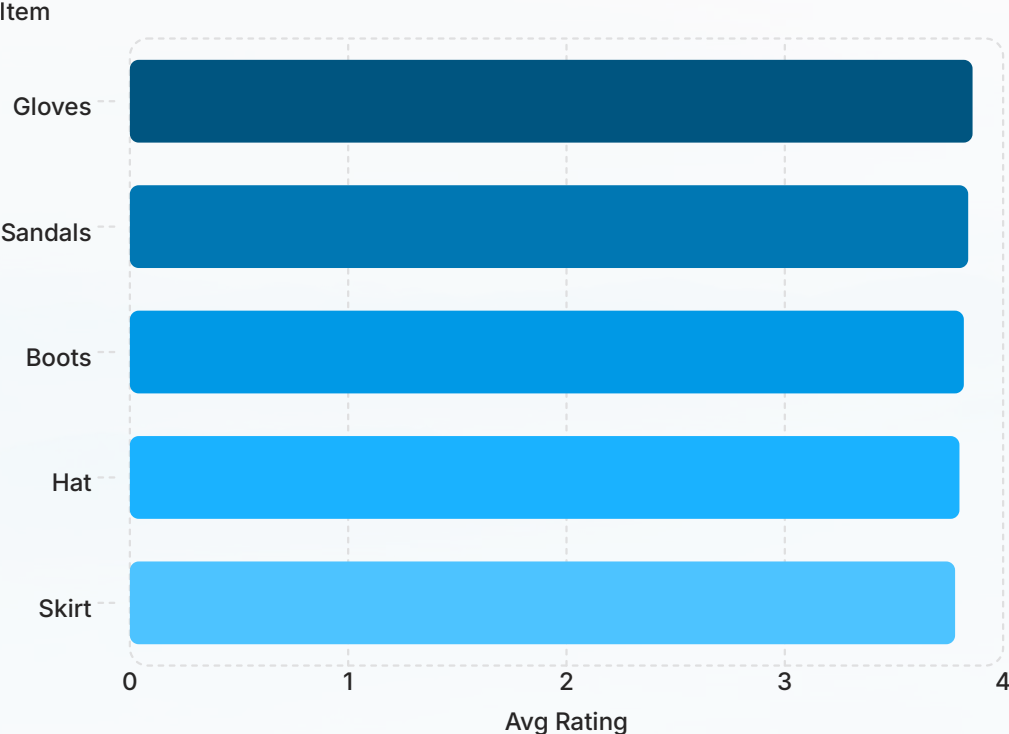


Key Findings

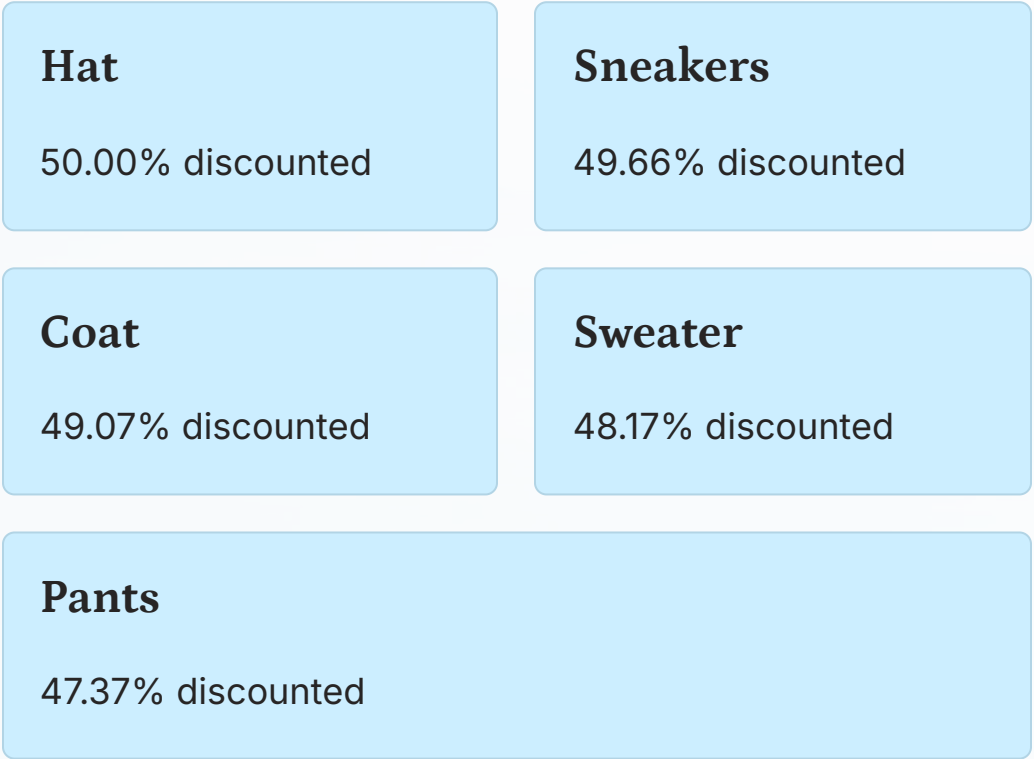
- **Male customers** generated 2× more revenue than females (\$157,890 vs \$75,191)
- **839 high-spending discount users** spent above average despite using discounts
- Express shipping averaged **\$60.48** vs Standard at **\$58.46**
- Subscribers (27% of customers) contributed **\$62,645** total revenue

Products, Ratings & Discounts

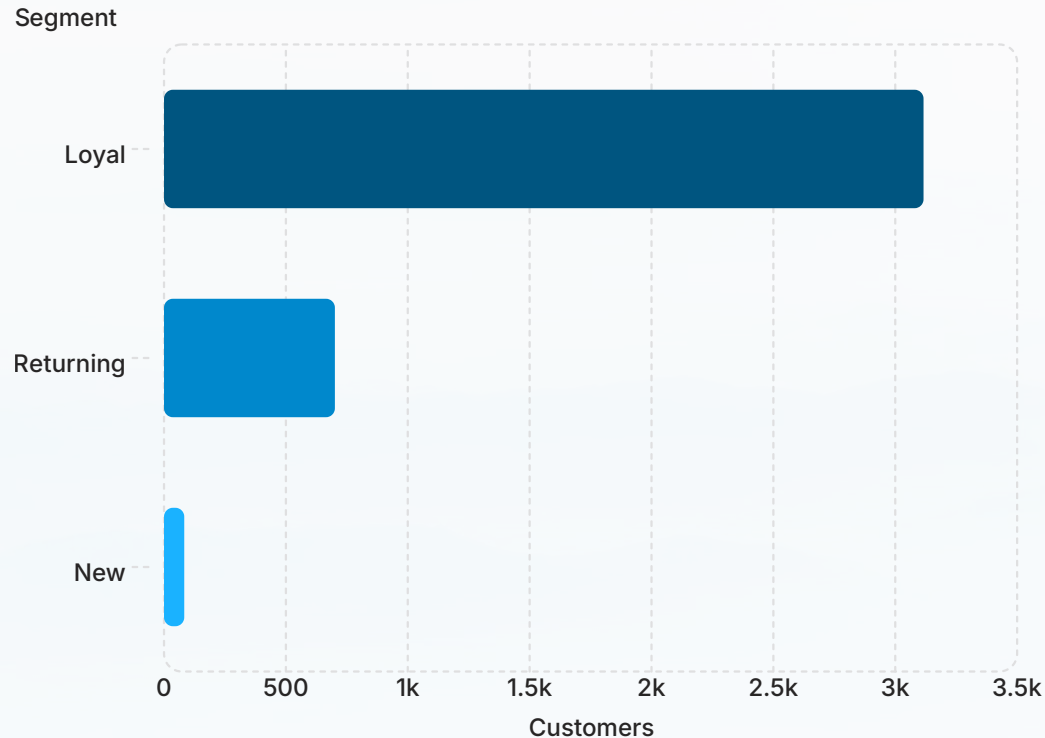
Top 5 Products by Rating



Most Discount-Dependent Products



Customer Segmentation



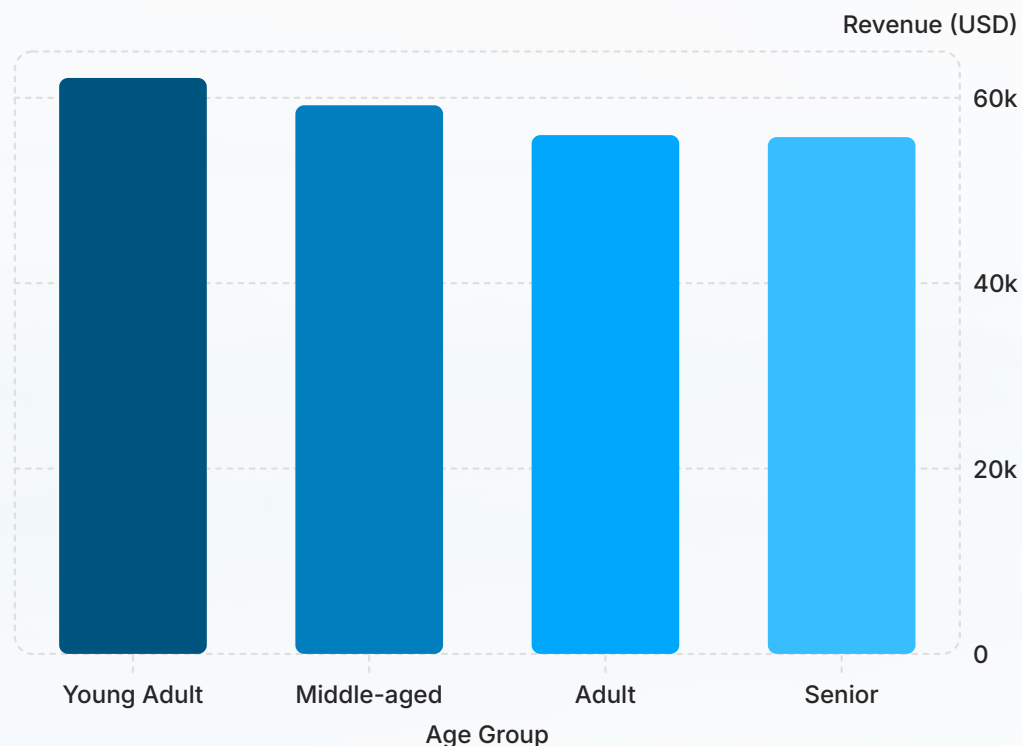
Segment Breakdown

Customers classified as **New**, **Returning**, or **Loyal** based on purchase history.

- **Loyal (80%)** — 3,116 customers with strong repeat purchase behavior
- **Returning (18%)** — 701 customers with some purchase history
- **New (2%)** — Only 83 first-time buyers

Of customers with 5+ purchases, 958 are subscribers vs 2,518 non-subscribers — repeat buyers show meaningful subscription uptake.

Revenue by Age Group



Age Group Insights

Revenue is relatively evenly distributed across age groups, with **Young Adults** leading at \$62,143 and **Seniors** contributing the least at \$55,763.

Top products by category: **Jewelry** (Accessories), **Blouse** (Clothing), **Sandals** (Footwear), and **Jacket** (Outerwear).

Interactive Power BI Dashboard

The Power BI dashboard provides a dynamic, at-a-glance overview of customer shopping behavior. It consolidates key metrics such as **3,900 total purchases**, an **average purchase value of \$59.76**, and an average review rating of **3.75**.

Visualizations include subscription status via a pie chart, revenue breakdown by product category and age group, and sales trends across different seasons. These actionable insights are crucial for refining marketing, product, and sales strategies to boost growth and customer engagement.

CUSTOMER BEHAVIOUR DASHBOARD

3.9K

No.of.customers

\$59.76

AVG Purchase Amount

3.75

AVG Review Rating

Subscription status

- ☐ No
☐ Yes

Gender

- ☐ Female
☐ Male

Category

- ☐ Accessories
☐ Clothing
☐ Footwear
☐ Outerwear

% OF CUSTOMERS BY SUBSCRIPTION STATUS



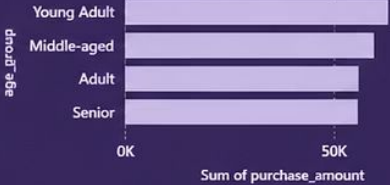
REVENUE BY CATEGORY



SALES BY CATEGORY



REVENUE BY AGE GROUP



SALES BY AGE GROUP



REVENUE BY SEASON



Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to grow the 27% base.



Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.



Review Discount Policy

Balance sales boosts with margin control on discount-heavy items.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users.